

BILMA, NigerWMO: **610170**Lat: **18.683N**Lon: **12.917E**Elev: **1171**StdP: **14.08**Time Zone: **1.00 (E01)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.8	43.2	17.5	13.9	75.7	18.5	14.6	75.0	33.8	61.6	27.0	62.4	0.1	100	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	28.1	111.7	70.3	110.3	70.2	108.8	70.0	77.3	98.9	76.1	98.4	75.2	98.1	4.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.6	121.9	88.5	69.7	114.0	88.9	68.0	107.4	89.9	41.6	97.0	40.4	98.2	39.4	97.9	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.2	21.0	17.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			30.4	80.9	1.9	3.6	29.1	83.5	28.0	85.5	26.9	87.5	25.6	90.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.1	62.4	69.6	77.8	86.4	92.4	95.2	94.7	93.6	91.6	83.3	73.0	64.9
	DBStd	12.60	5.34	6.25	6.25	6.00	3.63	3.97	2.98	3.08	3.14	5.33	5.15	5.46
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	212	116	21	1	0	0	0	0	0	0	0	3	71
	CDD50	11728	386	549	862	1093	1315	1356	1384	1352	1248	1031	690	462
	CDD65	6465	37	150	399	643	850	906	919	887	798	566	243	67
	CDH74	99337	615	1977	5098	9443	13403	14977	15209	14336	12344	7813	3130	992
	CDH80	66184	165	843	2787	6149	9414	10914	10857	9952	8440	4849	1514	300
Wind	WSAvg	4.4	5.1	5.5	6.1	5.5	4.3	3.8	3.4	3.5	3.2	4.3	3.7	5.0
Precipitation	PrecAvg	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.0	0.0	0.0	0.0
	PrecMax	1.6	0.0	0.0	0.0	0.6	0.5	0.2	0.8	1.2	0.2	0.2	0.0	0.2
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.4	0.0	0.0	0.0	0.1	0.1	0.1	0.2	0.3	0.0	0.0	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.5	97.0	103.5	110.2	113.1	114.2	111.9	110.7	109.7	105.7	97.8	90.3
		MCWB	60.7	62.7	65.8	68.4	69.7	70.4	71.6	72.8	71.7	68.3	64.4	60.9
	2%	DB	84.6	93.4	100.3	107.5	111.3	112.4	109.9	108.8	107.8	103.5	95.4	86.5
		MCWB	56.8	60.8	63.6	67.0	68.5	70.6	71.4	72.4	70.7	67.5	63.3	58.2
	5%	DB	80.8	89.4	97.2	105.4	109.4	110.7	108.3	106.8	106.4	101.2	92.3	83.6
		MCWB	54.9	59.1	62.8	66.0	67.8	70.3	71.4	72.7	70.2	67.0	62.0	57.5
	10%	DB	77.4	85.6	93.5	102.7	107.1	108.8	106.5	104.9	104.7	98.2	89.2	79.9
		MCWB	53.7	57.4	61.1	65.1	67.3	70.0	71.3	72.9	69.9	65.9	60.9	55.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.3	64.7	68.8	72.5	75.0	76.5	79.0	80.9	76.7	71.3	66.5	61.8
		MCDB	88.6	93.8	97.2	100.1	100.6	104.6	101.5	99.6	100.2	100.7	93.2	87.6
	2%	WB	57.8	61.8	65.8	69.7	72.6	75.1	76.5	77.9	75.2	69.1	64.4	59.6
		MCDB	81.5	89.8	95.3	98.9	100.0	103.1	99.7	98.3	100.3	99.2	92.3	84.4
	5%	WB	55.7	59.9	63.8	67.9	71.0	73.8	75.2	76.6	74.1	67.9	62.8	57.9
		MCDB	78.2	87.3	94.1	100.7	101.2	101.8	99.2	96.9	98.8	98.3	90.8	81.7
	10%	WB	54.2	58.0	62.1	66.2	69.3	72.7	74.2	75.7	72.8	66.5	61.2	56.3
		MCDB	76.0	84.5	91.6	99.1	102.1	101.1	98.3	96.4	97.2	96.5	87.7	78.6
Mean Daily Temperature Range	5% DB	MDBR	30.6	31.1	30.4	30.3	29.9	28.1	25.1	23.2	27.8	30.7	32.5	29.8
		MCDBR	34.7	34.9	34.0	32.1	33.8	30.1	27.3	25.5	29.6	31.1	34.5	34.1
		MCWBR	16.3	15.2	14.1	12.1	11.8	10.7	9.6	8.7	10.5	11.5	14.7	15.6
	5% WB	MCDBR	32.9	33.3	33.2	30.6	28.3	24.2	23.4	21.9	25.0	29.8	33.4	31.4
		MCWBR	16.4	15.4	14.6	12.9	11.2	9.6	8.8	8.0	9.5	12.0	14.9	15.0
Clear-Sky Solar Irradiance	taub		0.478	0.530	0.593	0.685	0.728	0.734	0.724	0.736	0.695	0.610	0.492	0.470
	taud		1.759	1.639	1.514	1.395	1.330	1.344	1.396	1.396	1.423	1.526	1.713	1.755
	Ebn at Noon		249	244	235	216	204	201	204	202	208	220	243	246
	Edh at Noon		67	80	93	106	112	109	104	105	101	88	70	65
All-Sky Solar Radiation	RadAvg		1706	1918	2164	2282	2358	2340	2296	2162	2133	1971	1740	1597
	RadStd		37	62	103	89	87	75	53	65	51	51	37	45